Math 210A Notes

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1 Basics

Proposition (Bijection)**.** The map f is a bijection if and only if there exists $g : B \to A$ such that $f \circ g$ is the identity map on B and $g \circ f$ is the identity map on A.

Definition (Binary Relations, Equivalence Relations, Equivalence Classes, Partitions)**.** (1) A **binary relation** on a set A is a subset R of $A \times A$ and we write $a \sim b$ if $(a, b) \in R$.

(2) The relation $\sim$ on A is said to be:

   (a) **Reflexive** if $a \sim a$ for all $a \in A$;

   (b) **Symmetric** if $a \sim b$, then $b \sim a$ for all $a, b \in A$;

   (c) **Transitive** if $a \sim b$ and $b \sim c$ implies $a \sim c$ for all $a, b, c \in A$.

Proposition (Isomorphism Properties)**.** If $\varphi : G \to H$ is an isomorphism, then

(a) $|G| = |H|$

(b) G is abelian if and only if H is abelian.

(c) for all $x \in G$, $|x| = |\varphi(x)|$.

2 Group Actions

Definition (Group Actions)**.** A **group action** of a group G on a set A is a map $G \times G \to A$ (written $g \cdot a$ for all $g \in G$ and $a \in A$) satisfying the following properties:

(1) $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a$ for all $g_1, g_2 \in G$ and $a \in A$;

(2) $1 \cdot a = a$ for all $a \in A$.

Example (Group Actions)**.** (1) The additive group $\mathbb{R}$ that acts on $\mathbb{R}^2$ by $r \cdot (x, y) = (x + ry, y)$.

(2) Let G be any group and let $A = G$. The map $g \cdot a : G \times A \to G$ defined by

$$g \cdot a = gag^{-1} \ \forall g, a \in G$$

satisfy the axioms of a group action.

2.1 Permutation Representation of a Group Action

Let the group G act on the set A. For each fixed $g \in G$, the map $\sigma_g : A \to A$ is a

(1) Permutation for each fixed $g \in G$ and

(2) The map from G to S_A defined by $g \mapsto \sigma_g$ is a homomorphism; that is, the map $\varphi : G \to S_A$ is any homomorphism from a group G to a symmetric group S_A defined by

$$g \cdot a = \varphi(g)(a).$$

3 Section 2.1: Subgroups

Definition (Subgroup). Let G be a group. The subset G of G is a **subgroup** of G if $H \neq \emptyset$ and H is closed under products and inverses; that is, for all $x, y \in H$ implies $x^{-1} \in H$ and $xy \in H$. If H is a subgroup of G, we shall write $H \leq G$.

Definition (The Subgroup Criterion). A subset H of a group G is a subgroup if and only if

(1) $H \neq \emptyset$, and

(2) For all $x, y \in H$, $xy^{-1} \in H$.

4 Section 2.2: Centralizers, Normalizers, Stabilizers, and Kernels

4.1 Centralizers, Centers, and Normalizers

Definition (Centralizer). Define $C_G(A) = \{g \in G : gag^{-1} = a \ \forall a \in A\}$. This subset of G is called the **centralizer** of A in G. Since $gag^{-1} = a$ if and only if $ga = ag$, $C_G(A)$ is the set of elements of G which commute with every element of A.

Definition (Center). Define $Z(G) = \{g \in G : gx = xg \ \forall x \in G\}$, the set of elements commuting with all elements of G. This subset of G is called the **center** of G.

Definition (Normalizer). Define $gAg^{-1} = \{gag^{-1} : a \in A\}$. Define the **normalizer** of A in G to be the set $N_G(A) = \{g \in G : gAg^{-1} = A\}$.

- If G is abelian if and only if $Z(G) = G$.
- G is abelian if and only if $C_G(A) = N_G(A)$ for any subset A of G.

4.2 Stabilizers and Kernels of Group Actions

In many cases, we can make our task of proving certain subsets of a group by making it about a problem about whether that subset is equal to either its stabilizer or kernel of a group action. The three main sets mentioned above are just special cases of group actions acting on either a fixed element of some set S or keeping all the elements of S fixed.

Definition (Stablizer). If G is a group acting on a set S and s is some fixed element of S, the **stabilizer** of s in G is the set

$$G_s = \{g \in G : g \cdot s = s\}.$$

Definition (Kernel). Let G be a group acting on a set S. Then the **kernel** of the action of G on S is the set
$$\{g \in G : g \cdot s = s \ \forall s \in S\}.$$

Let $S = \mathcal{P}(G)$ (the power set of G) (the collection of all subsets of G, and let G act on S by *conjugation*), that is, for each $g \in G$ and each $B \subseteq G$ let
$$g : B \to gBg^{-1} \text{ where } gBg^{-1} = \{gbg^{-1} : b \in B\}.$$
This action gives us the stabilizer of A in G; that is,
$$\begin{aligned} N_G(A) &= \{g \in G : gAg^{-1} = A\} \\ &= \{g \in G : g \cdot A = A\} \\ &= G_A \end{aligned}$$
which is clearly a subgroup of G.

5 Cyclic Groups and Cyclic Subgroups

Definition (Cyclic). A group H is **cyclic** if H can be generated by a single element; that is, there exists some element $x \in H$ such that $H = \{x^n : n \in \mathbb{Z}\}$ (where as usual the operation is multiplication).

- If the group in question is additive then we say that $H \subseteq G$ is cyclic if
$$H = \{nx : n \in \mathbb{Z}\}$$
- We say that H is generated by x (or x is the generator of H).
- A cyclic group may have more than one generator.

Proposition. If $H = \langle x \rangle$, then $|H| = |x|$ (where if one side of this equality is infinite, so is the other). More specifically,

(1) If $|H| = n < \infty$, then $x^n = 1$ and $1, x, x^2, \ldots, x^{n-1}$ are all distinct elements of H, and

(2) If $|H| = \infty$, then $x^n \neq 1$ for all $n \neq 0$ and $x^a \neq x^b$ for all $a \neq b$ in $\mathbb{Z}$.

Proposition ((3)). Let G be an arbitrary group, $x \in G$ and let $m, n \in \mathbb{Z}$. If $x^n = 1$ and $x^m = 1$, then $x^d = 1$, where $d = (m, n)$. In particular, if $x^m = 1$ for some $m \in \mathbb{Z}$, then $|x| \mid m$.

Theorem ((4)). Any two cyclic groups of the same order are isomorphic. More specifically,

(1) if $n \in \mathbb{Z}^+$ and $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n, then the map
$$\begin{aligned} \varphi : \langle x \rangle &\to \langle y \rangle \\ x^k &\mapsto y^k \end{aligned}$$
is well defined and is an isomorphism.

Proposition ((5))**.** Let G be a group, let $x \in G$ and let $a \in \mathbb{Z} - \{0\}$.

(1) If $|x| = \infty$, then $|x^a| = \infty$.

(2) If $|x| = n < \infty$, then $|x^a| = \dfrac{n}{(n,a)}$.

(3) In particular, if $|x| = n < \infty$ and a is a positive integer dividing n, then $|x^a| = \frac{n}{a}$.

Proposition ((6))**.** Let $H = \langle x \rangle$.

(1) Assume $|x| = \infty$. Then $H = \langle x^a \rangle$ if and only if $a = \pm 1$.

(2) Assume $|x| = n < \infty$. Then $H = \langle x^a \rangle$ if and only if $(a, n) = 1$. In particular, the number of generators of a H is $\varphi(n)$ (where φ is Euler's φ-function).

This is mainly used to give a description of all the subgroups of H that generate H.

Example (Multiplicative Group)**.** Find all the subgroups of $Z_{20} = \langle x \rangle$.

(1) Find all the divisors of 20. That is, the divisors of 20 are

$$\{1, 2, 4, 5, 10, 20\}.$$

(2) From the above proposition (2) the divisors of 20 gives us the unique subgroups of $\langle x \rangle$ that generate Z_{20}. These subgroups being

$$\langle x \rangle, \langle x^2 \rangle, \langle x^4 \rangle, \langle x^5 \rangle, \langle x^{10} \rangle, \langle x^{20} \rangle.$$

(3) Using prop (5), we can find the order of each subgroup:

$$\begin{aligned}
|\langle x^2 \rangle| &= \frac{20}{(2,20)} = \frac{20}{2} = 10 \text{ elements in } \langle x^2 \rangle \\
|\langle x^4 \rangle| &= \frac{20}{(4,20)} = \frac{20}{4} = 5 \text{ elements in } \langle x^4 \rangle \\
|\langle x^5 \rangle| &= \frac{20}{(5,20)} = \frac{20}{5} = 4 \text{ elements in } \langle x^5 \rangle \\
|\langle x^{10} \rangle| &= \frac{20}{(10,20)} = \frac{20}{10} = 2 \text{ elements in } \langle x^{10} \rangle \\
|\langle x^{20} \rangle| &= \frac{20}{(20,20)} = \frac{20}{20} = 1 \text{ element in } \langle x^{20} \rangle
\end{aligned}$$

(4) List out the generators of each subgroup by computing the Euler-Totient function $\varphi(n)$. For example, given the cyclic subgroup $\langle x^2 \rangle$ which has order 10, we have

$$\varphi(10) = \varphi(5 \cdot 2) = 4 \text{ generators.}$$

Note that φ gives us the number of $a \in \mathbb{Z}^+$ up to 10 such that $(a, 10) = 1$ (where a and 10 are relatively prime). Indeed, we have

- $(1, 10) = 1 \implies (x^2)^1 = x^2$ is a generator;
- $(2, 10) = 2 \implies (x^2)^2 = x^4$ is NOT a generator;
- $(3, 10) = 1 \implies (x^2)^3 = x^6$ is a generator;
- $(4, 10) = 2 \implies (x^2)^4 = x^8$ is NOT a generator;

- $(5, 10) = 5 \implies (x^2)^5 = x^{10}$ is NOT a generator;
- $(6, 10) = 2 \implies (x^2)^6 = x^{12}$ is NOT a generator;
- $(7, 10) = 1 \implies (x^2)^7 = x^{14}$ is a generator;
- $(8, 10) = 2 \implies (x^2)^8 = x^{16}$ is NOT a generator;
- $(9, 10) = 1 \implies (x^2)^9 = x^{18}$ is a generator;
- $(10, 10) = 10 \implies (x^2)^{10} = x^{20}$ is NOT a generator;

(5) To describe the containments of each of these subgroups in relation to each other, we fix each divisor in our list and identify which ones can be divided by it. For instance, we see that

$$\begin{aligned} 1 &\mid 2, 4, 5, 10, 20 \\ 2 &\mid 4, 10, 20 \\ 4 &\mid 20 \\ 5 &\mid 10, 20 \\ 10 &\mid 20. \end{aligned}$$

This gives us the following containments:

$$\begin{aligned} &\langle x^{20}\rangle \subseteq \langle x^{10}\rangle \subseteq \langle x^5\rangle \subseteq \langle x\rangle \\ &\langle x^{20}\rangle \subseteq \langle x^{10}\rangle \subseteq \langle x\rangle \\ &\langle x^{20}\rangle \subseteq \langle x^4\rangle \subseteq \langle x^2\rangle \subseteq \langle x\rangle \\ &\langle x^{20}\rangle \subseteq \langle x^2\rangle \subseteq \langle x\rangle \\ &\langle x^{20}\rangle \subseteq \langle x\rangle. \end{aligned}$$

Theorem. Let $H = \langle x\rangle$ be a cyclic group.

(1) Every subgroup of H is cyclic. More precisely, if $K \leq H$, then either $K = \{1\}$ or $K = \langle x^d\rangle$, where d is the smallest positive integer such that $x^d \in K$.

(2) If $|H| = \infty$, then for any distinct nonnegative integers a and b, $\langle x^a\rangle \neq \langle x^b\rangle$. Furthermore, for every integer m, $\langle x^m\rangle = \langle x^{|m|}\rangle$, where $|m|$ denotes the absolute value of m, so that the nontrivial subgroups of H correspond with the integers $1, 2, 3, \ldots$.

(3) If $|H| = n < \infty$, then for each positive integer a dividing n there is a unique subgroup of H of order a. This subgroup is the cyclic group $\langle x^d\rangle$, where $d = \frac{n}{a}$. Furthermore, for every integer m, $\langle x^m\rangle = \langle x^{(n,m)}\rangle$ so that the subgroups of H correspond bijectively with the positive divisors of n.

6 Week 6

6.1 Cyclic

Proposition (8)**.** If $\mathcal{A}$ is any collection of subgroups of a group G, then $\bigcap_{H\in\mathcal{A}} H$ is a subgroup of G.

Proof. This can be shown pretty easily. ■

Definition (The Generating Set of a Group). If A is any subset of the group G define

$$\langle A \rangle = \bigcap_{A \subseteq H, H \leq G} H$$

where A is called the **generating set**.

Notice that in the notation of the above proposition, we have

$$\mathcal{A} = \{H \leq G : A \subseteq H\}.$$

Notice that $\mathcal{A} \neq \emptyset$ as G is an element of $\mathcal{A}$ since $G \leq G$ and $A \subseteq G$.

Proposition. $\langle A \rangle$ is the unique minimal element of $\mathcal{A}$.

Proof. Since $A \subseteq H$ for all $H \in \mathcal{A}$, $A \subseteq \langle A \rangle$. Then $\langle A \rangle \in \mathcal{A}$. Let $K \in \mathcal{A}$. Note that

$$\bigcap_{H \in \mathcal{A}} H \leq K.$$

That is, $\langle A \rangle \leq K$. Hence, $\langle A \rangle$ is the minimal element with respect to inclusion. ■

When A is finite, that is,

$$A = \{\alpha_1, \alpha_2, \ldots, \alpha_n\} \text{ for some } n \in \mathbb{N},$$

we write

$$\langle A \rangle = \langle a_1, a_2, \ldots, a_n \rangle.$$

This is a more concrete version of the previous set $\langle A \rangle = \bigcap_{H \leq G, K \subseteq H} H$. Define the following set

$$\overline{A} = \{a_1^{\varepsilon_1} a_2^{\varepsilon_2} a_3^{\varepsilon_3} \ldots a_n^{\varepsilon_n} : n \in \mathbb{N}, \varepsilon_i = \pm 1 \ \forall i\}.$$

If $A = \emptyset$, define $\overline{A} = \{1\}$.

Proposition. $\langle A \rangle = \overline{A}$

6.2 Quotient Groups and Homomorphisms

Let G be a group and $N \leq G$. Define a relation on G by

$$a \sim b \iff a^{-1}b \in N.$$

It is straightforward to prove that this defines an equivalence on relation on G. For $a \in G$, the equivalence class of a is

$$\begin{aligned}
\{b \in G : a \sim b\} &= \{b \in G : a^{-1}b \in N\} \\
&= \{b \in G : a^{-1}b = n \ \text{ for some } n \in N\} \\
&= \{b \in G : b = an \text{ for some } n \in N\} \\
&= \{an : n \in N\} \\
&:= aN.
\end{aligned}$$

Definition (Cosets). Suppose $N \leq G$ where G is a group and $g \in G$. Let

$$gN = \{gn : n \in N\}$$
$$Ng = \{ng : n \in N\}$$

be called a **left coset** and **right coset** of N in G, respectively. Any element of a coset is called a

representative of a coset.

We will denote the set of all left cosets of N (or right cosets) in G by G/N (read as G mod N).

Proposition (4). Let $N \leq G$ be a group and G/N forms a partition of G. For all $a, b \in G$, $aN = bN$ if and only if $a^{-1}b \in N$. In particular, $aN = bN$ if and only if a and b are representatives of the same coset.

Proof. Since we have recognized left cosets as equivalence classes induced by an equivalence relation, they form a partition, that is,

$$G = \bigcup_{g \in G} gN$$

and $g_1N = g_2N$ if and only if $g_1N \cap g_2N \neq \emptyset$ for all $g_1, g_2 \in G$.

($\Longrightarrow$) Suppose $a^{-1}b \in N$, then

$$a^{-1}b = n \text{ for some } n \in N.$$

It follows that $b = an \in aN$. Since N is a subgroup, $1 \in N$. Hence, $b \cdot 1 \in bN$. It follows that

$$aN \cap bN \neq \emptyset \Longrightarrow aN = bN.$$

($\Longleftarrow$) Suppose that $aN = bN$. Then $an = b$ for some $n \in N$. Hence, $n = a^{-1}b \in N$. We have

$$\begin{aligned} an = bn &\Longleftrightarrow a^{-1}b \in N \\ &\Longleftrightarrow b \in aN \text{ and } a \in aN \\ &\Longleftrightarrow a \text{ and } b \text{ are representatives of } aN \text{ (or } bN). \end{aligned}$$

■

Proposition (5). (1) The operation on G/N described by $aN \cdot bN = (ab)N$ for all $a, b \in G$ is well defined if and only if $gng^{-1} \in N$ for all $g \in G$ and $n \in N$.

(2) If the operation above is well-defined, then G/N forms a group, where $1N$ is the identity and $(gN)^{-1} = g^{-1}N$.

Proof. (1) ($\Longleftarrow$) (We will show that the operation defined above is well-defined) To show that the operation is well-defined, we need to show that

$$(ab)N = (a_1b_1)N.$$

Suppose that $gng^{-1} \in N$ for all $g \in G$ and $n \in N$. Let $a, a_1 \in aN$ and $b, b_1 \in bN$. Then $a_1 = an$ and $b_1 = bm$ for some $n, m \in N$. To get our result, we need to show that $a_1b_1 \in (ab)N$ if and only if $(a_1b_1)N = (ab)N$.

Now, we will prove the former. Indeed, we have

$$\begin{aligned} a_1b_1 &= (an)(bm) \\ &= a(bb^{-1})nbm \\ &= ab(b^{-1}nb)m. && (b^{-1}n(b^{-1})^{-1} \in N)) \end{aligned}$$

Hence, it follows that $a_1b_1 = (ab)n_1m$ where $n_1 \in N$ and $n_1 = b^{-1}n(b^{-1})^{-1}$.

($\Longrightarrow$) This direction is left for the reader to prove.

(2) This is left for the reader to prove. ■

7 Week 7

Last time we constructed cosets from an equivalence relation and looked at the set of all left cosets of subgroup N of a group G, denoted by

$$G/N = \{gN : g \in G\}.$$

We showed that the operation on G/H defined by

$$(aH)(bH) = (ab)N$$

is well-defined when $gng^{-1} \in H$ for all $g \in G$ and for all $h \in N$. This makes G/N a group. We will prove that this operation is associative, the rest is left to the reader.

Let $N \leq G$ a group where "coset multiplication" is well-defined. Let $aN, bN, cN \in G/N$ $(a, b, c \in G)$. Then

$$\begin{aligned} aN(bNcN) &= aN((bc)N) \\ &= (a(bc))N \\ &= ((ab)c)N \\ &= ((ab)N)cN \\ &= (aNbN)cN. \end{aligned}$$

Similarly, we can show that $1N$ is the identity of G/N and show that

$$(aN)^{-1} = a^{-1}N \quad \forall aN \in G/N.$$

Indeed, for every $gN \in G/N$

$$gN = (1_G g)N = (1_G N) \cdot (gN)$$

and

$$gN = (g1_G)N = (1_G N) \cdot (gN).$$

Now, we have that since $g \in G$ and $g^{-1} \in G$, it follows that

$$1_G N = (gg^{-1})N = (gN) \cdot (g^{-1}N)$$

and

$$1_G N = (g^{-1}g)N = (g^{-1}N) \cdot (gN).$$

This tells us that G/N is a group where N has the nice property of being "normal". This will be expanded in the following definition:

Definition (Normal Subgroups of G). A subgroup N of G is called **normal** in G if every element of g normalizes N. That is, N is normal in G if

$$gNg^{-1} = N \quad \forall g \in G.$$

If N is a normal subgroup of G, then we write $N \trianglelefteq G$.

Theorem (6). Let N be a subgroup of G. The following are equivalent

(1) $N \trianglelefteq G$

(2) $N_G(N) = G$

(3) $gN = Ng$ for all $g \in G$

(4) The operation coset multiplication is well-defined.

(5) $gNg^{-1} \subset N$ for all $g \in G$.

Why does (5) only need containment? It is because you can construct a bijective map between gNg^{-1} and N.

Example (Checking if $\langle s \rangle$ is normal in D_16)**.** We need to examine gsg^{-1} for an arbitrary $g \in D_{16}$. Note that

$$D_{16} = \{1, r, \dots, r^7, s, sr, \dots, sr^7\}.$$

Letting $g = s^i r^j$ where $i \in \{0, 1\}$ and $j \in \{0, 1, 2, \dots, 7\}$. Then we have

$$\begin{aligned} gsg^{-1} &= (s^i r^j)s(s^i r^j)^{-1} && (s^i = s^{-i}) \\ &= s^i r^j s r^{-j} s^i \end{aligned}$$

Note that when $i = 0$, we have

$$\begin{aligned} gsg^{-1} &= r^j s r^{-j} \\ &= r^j r^j s \\ &= r^{2j} s. \end{aligned}$$

When $j = 1$, this gives us $gsg^{-1} = r^2 s$. But note that $r^2 s \notin \langle s \rangle$ because otherwise r^2 is either the identity or s which r^2 is neither of these two which is a contradiction. Hence, gsg^{-1} cannot be in $\langle s \rangle$ and so it cannot be a normal subgroup of D_16.

Theorem (Big!)**.** A subgroup N of a group G is normal if and only if it is the kernel of some homomorphism.

Proof. ($\Longrightarrow$) Suppose $N \trianglelefteq G$. Let's define the mapping $\pi : G \to G/N$ by $\pi(g) = gN$ for all $g \in G$. Notice that since N is a normal subgroup of G, this map is well-defined. Let $g_1, g_2 \in G$. Then

$$\pi(g_1 g_2) = (g_1 g_2)N = (g_1 N)(g_2 N) = \pi(g_1)\pi(g_2).$$

Hence, π is a homomorphism. It remains to show that $\ker(\pi) = N$. Indeed, we have that

$$\begin{aligned} \ker(\pi) &= \{g \in G : \pi(g) = 1N\} \\ &= \{g \in G : gN = 1N\} \\ &= \{g \in G : g \in 1N\} \\ &= \{g \in G : g \in N\} \\ &= N. \end{aligned}$$

Hence, $\ker(\pi) = N$.

($\Longleftarrow$) This direction will be on homework. ■

Definition (Natural Projection)**.** Let N be normal in G. The homomorphism π from the previous theorem is called the **natural projection** (homomorphism) of G onto G/N. If $\overline{G} \le G/N$, the complete pre-image of $\overline{H}$ is $\pi^{-1}(\overline{H})$.

Remark. If $\overline{H} \subseteq G/N$, then $N \le \pi^{-1}(\overline{H})$ since

$$1N \in \overline{H} \Longrightarrow N = \ker(\pi) = \pi^{-1}(1N) \subseteq \pi^{-1}(\overline{H}).$$

Example (Complete Pre-image of Q_8). $\pi^{-1}(\overline{H}) = \{1, -1, i, -i\}$ where

$$\pi(i) = i\langle -1\rangle$$
$$\pi(-1) = -1\langle -1\rangle = 1\langle -1\rangle$$

7.1 More on Normal Subgroups

There are a lot of ways to see if a subgroup is normal.

Remark (Important). Let G be a group. Observe that $\{1\} \trianglelefteq G$ and $G \trianglelefteq G$ and $G/\{1\} \cong G$ and $G/G \cong \{1\}$.

- When G is clearly an additive group, we denote left and right cosets $g + N$ and $N + g$, respectively where $N \leq G$.
- When G is abelian, every subgroup is normal; that is, $N \leq G$ where

$$\forall n \in \mathbb{N} \quad gng^{-1} = gg^{-1}n = n \in \mathbb{N} \ \ \forall g \in G.$$

7.2 More on Cosets/Lagrange's Theorem

We move away from normal subgroups and just analyze subgroups.

Theorem (Lagrange's Theorem). If G is a finite group, and H is a subset of G, then

$$|H| \mid |G|$$

and the number of left cosets of H in G is $\frac{|G|}{|H|}$.

The idea of the proof of this theorem is as follows: (see problem 18 and 19 from section 1.7 in the book)

Proof. Recall that left cosets form a partition of G; that is,

$$G = \bigcup_{g \in G} gH.$$

There is a bijection from H to gH and is defined by $h \mapsto gh$ and so

$$|H| = |gH|.$$

At this point, we can conclude that $|G| = k|H|$ where k is the number of distinct left-cosets of H in G. This gives us

$$k = \frac{|G|}{|H|}.$$

∎

Definition (Index of a Subgroup). If G is a group (possibly infinite) and H is a subgroup of G, then the number of distinct left cosets of H in G is called the **index of H in G** which is denoted by $|G : H|$ (or $[G : H]$).

Corollary (Corollary to Lagrange's Theorem). If G is a finite group and $x \in G$, then $|x| \mid |G|$.

Proof. We proved that $|x| = |\langle x\rangle|$ where $\langle x\rangle \leq G$. The rest follows from Lagrange's Theorem. ■

Remark. For a finite group G with $H \leq G$, $|G : H| = \frac{|G|}{|H|}$.

Example. Let $G = \mathbb{Z}$ and let $H = 3\mathbb{Z}$, then $|\mathbb{Z} : 3\mathbb{Z}|$ is the number of left cosets of $3\mathbb{Z}$ in $\mathbb{Z}$. Recall that

$$3\mathbb{Z} = \{3x : x \in \mathbb{Z}\}$$

where

$$\begin{aligned} 0 + 3\mathbb{Z} &= 3\mathbb{Z} \\ 1 + 3\mathbb{Z} &= \{1 + 3x : x \in \mathbb{Z}\} \\ 2 + 3\mathbb{Z} &= \{2 + 3x : x \in \mathbb{Z}\} \\ 3 + 3\mathbb{Z} &= 0 + 3\mathbb{Z}. \end{aligned}$$

That is,

$$|\mathbb{Z} : 3\mathbb{Z}| = 3 = |\mathbb{Z}/3\mathbb{Z}|.$$

Corollary. If G is a group of prime order p, then G is cyclic.

Proof. Let $x \in G$ where $x \neq 1$. Then $|x| \mid |G|$ since $|G| = p$ a prime. Hence, $|x| \in \{1, p\}$. Since $x \neq 1$, we have $|x| \neq 1$ and so we would need to have $|x| = p$ and so $\langle x\rangle = G$. ■

Example. A subgroup H of a group G with index $|G : H| = 2$, is normal.

Proof. Let $g \in G - H$, then $gH \neq 1H$. Since $|G : H| = 2$, there are two distinct left cosets of H in G and since one of them is $1H$, the other must be gH. Similarly, there are only two distinct right cosets of H in G; namely, $H1$ and Hg. Since $1H = H1$ and cosets form a partition of G, we have that $gH = G - H = Hg$. Hence, the left and right cosets of H are equal and so H is normal in G. ■

Example (Normal Subgroups of is NOT transitive)**.** Note that the relation "is a normal subgroup of" is not transitive. Let $G = D_8$. Note that the $|D_8| = 8$ and $|\langle s\rangle| = 2$, $|\langle s, r^2\rangle| = 4$. Clearly, we have

$$\langle s\rangle \subseteq \langle s, r^2\rangle \subseteq D_8.$$

We have

$$|D_8 : \langle s, r^2\rangle| = 2$$

and $|\langle s, r^2\rangle : \langle s\rangle| = 2$. Thus,

$$\langle s\rangle \trianglelefteq \langle s, r^2\rangle \trianglelefteq D_8$$

but $\langle s\rangle$ is NOT normal in D_8. Note that $rs \neq sr$ but $rs = sr^{-1}$ where r^2s is not 1 and not s.

Definition. Let $H, K \leq G$ where G is a group. Define

$$HK = \{hk : h \in H,\ k \in K\}.$$

Theorem (Big Theorem of this lecture). If H and K are finite subgroups of a group, then

$$|HK| = \frac{|H||K|}{|H \cap K|}$$

where HK need not be a group for the above to hold.

Proof. Notice that HK is the union of left cosets of K; that is,

$$HK = \bigcup_{h \in H} hK.$$

Since each coset of K has $|K|$ elements, we will count the number of distinct cosets in the above union. We know $h_1k = h_2k$ for $h_1, h_2 \in H$ if and only if $h_2^{-1}h_1 \in K$. It follows that

$$\begin{aligned} h_1k = h_2k &\iff h_2^{-1}h_1 \in K \cap H \\ &\iff h_1(K \cap H) = h_2(K \cap H). \end{aligned}$$

Thus, the number of distinct cosets of the form hK with $h \in H$ is the same as the number of distinct cosets of $K \cap H$ in H. Since we have that $K \cap H$ is a subgroup of H, this is $\frac{|H|}{|K \cap H|}$. Therefore, HK consists of $\frac{|H|}{|H \cap K|}$ distinct cosets of K of which contains $|K|$ elements, it follows that

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

■

Example (An example where HK is not a subgroup).

8 Week 8

8.1 Monday

Recall that a lot of the problems given in the homework are based on more general theory of the results that we have been going thus far. For instance, one can prove that If G is cylic and φ is a homomorphism from G to H, then $\varphi(G)$ is cyclic. Note that $\pi : G \to G/N$ where $g \mapsto gN$ and π is a homomorphism. Another problem we can do is prove that if G is cyclic, then G/N is cyclic.

For an $\psi : G \to H$ where ψ is a group homomorphism. If $|x| = n < \infty$, then $|\psi(x)| | |x|$. Because π is a homomorphism from $G \to G/N$. Hence, the homework problem 2 follows immediately.

Last time we saw that for $H, K \leq G$ we have

$$|HK| = \frac{|K||H|}{|K \cap H|}.$$

Note that HK may not always be a subgroup of G.

Example. Let $S_3 = G$ and $H = \langle(1\ 2)\rangle$ and $K = \langle(2\ 3)\rangle$. Then

$$|HK| = \frac{|H||K|}{|H \cap K|} = \frac{2 \cdot 2}{1} = 4.$$

However, we see from Lagrange's theorem that if $HK \leq G$, then $4 \mid 3!$ which is a contradiction. We can further deduce that

$$\langle(1\ 2), (1\ 3)\rangle = S_3.$$

Since $4 \leq |\langle(1\ 2), (2\ 3)\rangle| \leq 6$ and must also divide 6. So, (1 2) and (2 3) generates all of S_3.

Proposition. If H, K are subgroups of a group G, $HK \leq G$ if and only if $HK = KH$. *Note that $HK = KH$ does not indicate that the elements of the set commute with each other. Only that $hk \in HK$, we have that $hk = k_1h_1$ for some $h_1 \in H$ and $k_1 \in K$*

Proof. Assume that $HK = KH$. Our goal is to show that HK is a subgroup of G. Since H and K are non-empty, then $HK \neq \emptyset$. It remains to show that given $a, b \in HK$ that $ab^{-1} \in HK$. To this end, let $a, b \in HK$. Then $a = h_1k_1$ and $b = h_2k_2$. Then

$$\begin{aligned} ab^{-1} &= (h_1k_1)(h_2k_2)^{-1} \\ &= (h_1k_1)(k_2^{-1}h_2^{-1}) \end{aligned}$$

Since $K \leq G$, we have $k_1k_2^{-1} = k_3 \in K$ and $h_2^{-1} = h_3 \in H$. This gives that

$$ab^{-1} = h_1k_3h_3.$$

Since $HK = KH$, we know that $k_3h_3 \in HK$. That is, $k_3h_3 = h_4k_4$ for some $h_4 \in H$ and $k_4 \in K$. So,

$$ab^{-1} = h_1h_4k_4$$

and letting $h_1h_4 = h_5$ which is in H by closure, we have that

$$ab^{-1} = h_5k_4 \in HK.$$

Conversely, suppose that $HK \leq G$. Our goal is to show that $HK = KH$; that is, $HK \subseteq KH$ and $KH \subseteq HK$. Since $K \leq HK$ and $H \leq HK$ and so $KH \leq HK$ by closure of HK. To show that $HK \leq KH$, let $hk \in HK$. Since HK is a subgroup, then hk is the inverse of some element $a \in HK$. That is, $hk = a^{-1} = (h_1k_1)^{-1} = k_1^{-1}h_1^{-1} \in KH$. It follows that $HK \leq KH$. Furthermore, $HK = KH$. ■

Example. Let $G = D_8$, $H = \langle r\rangle$ and $K = \langle s\rangle$. Notice that $rs \in HK$ and $rs = sr^{-1} \in KH$. By the way,

$$|HK| = \frac{|H||K|}{|H \cap K|} = \frac{4 \cdot 2}{1} = 8.$$

So, $HK = D_8 = KH$.

Corollary. If H and K are subgroups of a group G and $H \leq N_G(K)$, then HK is a subgroup of G. In particular, if $K \trianglelefteq G$, then HK is a subgroup of G for all $H \leq G$. *Note: If $K \trianglelefteq G$, then $N_G(K) = G$ and certainly $H \subseteq N_G(K)$ for any $H \leq G$*

Proof. We will prove that $HK = KH$. Let $h \in H$ and $k \in K$. By assumption, we have that

$h \in N_G(K)$; that is, $hkh^{-1} \in K$. Hence, $hk = hk(h^{-1}h)$ where $hkh^{-1} \in K$ and $h \in H$. Thus, $hk \in KH$. Thus, $HK \subseteq KH$. Similarly, we want to show that $kh = hh^{-1}kh$ where $h \in H$ and $h^{-1}kh \in K$ since $h \in N_G(K)$ implies that $h^{-1} \in N_G(K)$. It follows that $KH \subseteq HK$ and hence $HK = KH$. ■

Theorem. Let H, K be subgroups of a group G with $H \leq K \leq G$. Then the index

$$|G : H| = |G : K||K : H|.$$

Proof. Let g_i be a distinct representative for a left coset of H in G for all $i \in I$ where I is some indexing set. So, we have

$$\{g_iH : i \in I\} = G/H = \{gH : g \in G\}.$$

We have $g_iH = g_jH$ if and only if $g_i = g_j$ (we are counting the elements of gH because we don't know if it will be finite or infinte). *At this point we are invoking the axiom of choice.* Let $\psi : I \times K/H \to G/H$ by

$$\psi(i, kH) = g_ikH.$$

To see that this is well-defined, let us suppose that $k_1H = k_2H$ for some k_1 and k_2 in K. That is, $k_1^{-1}k_2 \in H$. Then $\psi(i, k_1H) = g_ik_1H$ and $\psi(i, k_2H) = g_ik_2H$. So,

$$\begin{aligned}(g_ik_1)^{-1}(g_ik_2) &= (k_1^{-1}g_i^{-1})(g_iK_2)\\ &= k_1^{-1}k_2 \in H\end{aligned}$$

by assumption. Hence, ψ is well-defined. We now show that ψ defines a bijection. Suppose that

$$\begin{aligned}&\psi(i, k_1H) = \psi(j, k_2H)\\ &\implies (g_ik_1)H = (g_jk_2)H\\ &\implies (g_ik_1)^{-1}(g_jk_2) \in H.\end{aligned}$$

This implies that $k_1^{-1}g_i^{-1}g_jk_2 = h$ (call this (*)) for some $h \in H$. This tells us that $g_i^{-1}g_j = k_1hk_2^{-1}$ for some $h \in H$. Note that $H \leq K \leq G$ and so $g_i^{-1}g_j \in K$ since $H \subseteq K$. Then $g_iK = g_jK$. Thus, we have $g_i = g_j$. This further implies that $i = j$. Using this in (*) gives

$$\begin{aligned}&k_1^{-1}h_i^{-1}g_jk_2 = h\\ &\implies k_1^{-1}k_2 = h \in H\\ &\implies k_1H = k_2H.\end{aligned}$$

Hence, we see that φ is one-to-one. Let $gH \in G/H$. Since the left cosets of k partition the group G, we have that $g \in g_iK$. That is, $g = g_ik$ for some $k \in K$. Hence, $\varphi(i, kH) = g_ikH = gH$. Thus, ψ is surjective. Hence,

$$\underbrace{|G : H|}_{G/H} = \underbrace{|G : K|}_{I} \cdot \underbrace{|K : H|}_{K/H}.$$

■

Theorem (First Isomorphism Theorem). If $\varphi : G \to H$ is a homomorphism of groups, then

$$\ker(\varphi) \trianglelefteq G,$$

and

$$G/\ker(\varphi) \cong \varphi(G).$$

Proof. *Proof is left to the reader.* Take $\mu : G/\ker(\varphi) \to \varphi(G)$ where $gK \mapsto \varphi(g)$ and prove that is well-defined and is a bijection. ■

End of Monday Lecture

8.2 Wednesday

Recall that the main goal for this subject is to be able to classify groups. One way we have been gaining information about a certain group is by multiplying two subgroups together. In general, we are trying to develop tools to be able to better classify certain groups.

Theorem (First Isomorphism Theorem)**.** If $\varphi : G \to H$ is a homomorphism of groups, then

$$\ker(\varphi) \trianglelefteq G,$$

and

$$G/\ker(\varphi) \cong \varphi(G).$$

Proof. *Proof is left to the reader.* Take $\mu : G/\ker(\varphi) \to \varphi(G)$ where $gK \mapsto \varphi(g)$ and prove that is well-defined and is a bijection. ■

Corollary (17 from book)**.** Let $\varphi : G \to H$ be a group homomorphism

(1) φ is injective iff $\ker(\varphi) = 1$

(2) $|G : \ker(\varphi)|$

Theorem (Second Isomorphism Theorem)**.** Let G be a group, let A and B be subgroups of G. Also assume that A is contained in the normalizer $N_G(B)$. Then $AB \leq G$ and B is normal in AB, $A \cap B$ is normal in A, and

$$AB/B \cong A/(A \cap B).$$

(The second isomorphism theorem is often called the Diamond Isomorphism Theorem)

Proof. Since $A \leq N_G(B)$ implies $AB \leq G$ and $B \leq N_G(B)$, and $B \leq AB$, it follows that $AB \leq N_G(B)$, and so $B \trianglelefteq AB$. Since $B \trianglelefteq AB$, then AB/B is a group. Define $\varphi : A \to AB/B$ by $\varphi(a) = aB$ for all $a \in A$. Let $a_1, a_2 \in A$. Hence,

$$\varphi(a_1a_2) = (a_1a_2)B = a_1Ba_2B = \varphi(a_1)\varphi(a_2)$$

since AB/B is a group. Notice that $abB = aB$ since $b \in B$. That is, for all $abB \in AB/B$

$$abB = aB.$$

Hence, our mapping is surjective. Hence, $\varphi(A) = AB/B$. Notice that

$$\begin{aligned}\ker(\varphi) &= \{a \in A : \varphi(a) = 1B\}\\ &= \{a \in A : aB = 1B\}\\ &= \{a \in A : a \in B\}\\ &= A \cap B.\end{aligned}$$

Now, we have $\varphi(A) = AB/B$ and $\ker(\varphi) = A \cap B$. Thus, by the first isomorphism theorem, we have

$$A/\ker(\varphi) \cong \varphi(A) \Longrightarrow A/(A \cap B) \cong AB/B$$

where $A \cap B \trianglelefteq A$. ■

Theorem (Third Isomorphism Theorem)**.** Let G be a group, and let H and K be normal subgroups of G with $H \leq K$. Then

$$(G/H)/(K/H) \cong G/K.$$

Proof. Since $K \trianglelefteq G$, we have $\pi_H(K) = \{kH : k \in K\}$, we know that

$$\pi_H(K) \trianglelefteq \pi_H(G) = G/H.$$

Notice that

$$\{kH : k \in K\}$$

Now, let's define our mapping $\varphi : G/H \longrightarrow G/K$ by $gH \mapsto gK$. Now, we want to show that φ is well-defined. Suppose that $g_1H = g_2H$ if and only if $g_1 = g_2h$ for some $h \in H$. Our goal is to show that $\varphi(g_1H) = \varphi(g_2H)$; that is, we want to show that $g_1K = g_2K$. Since $H \leq K$, so $h \in K$ as well; that is, $g_1 = g_2h$ where $h \in K$. So, $g_1K = g_2K$. Hence, $\varphi(g_1H) = \varphi(g_2H)$ and our mapping is well-defined. Next, we will show that this mapping is a homomorphism. Let $g_1H, g_2H \in G/H$ and

$$\varphi(g_1Hg_2H) = \varphi(g_1g_2H) = g_1g_2K = (g_1K)(g_2K) = \varphi(g_1H)\varphi(g_2H).$$

Hence, φ is a homomorphism which is clearly surjective. Hence,

$$\varphi(G/H) = G/K.$$

Finally, we will show that $\ker(\varphi) = K/H$. Indeed, we see that

$$\begin{aligned}
\ker(\varphi) &= \{gH \in G/H : \varphi(gH) = 1K\} \\
&= \{gH \in G/H : gK = 1K\} \\
&= \{gH \in G/H : g \in K\} \\
&= K/H.
\end{aligned}$$

Therefore, by the first isomorphism theorem,

$$(G/H)/(K/H) \cong G/K$$

where $\varphi(G/H) = G/K$ and $\ker(\varphi) = K/H$. ■

The next theorem tell us that when there is a normal subgroup of G, there exists a bijection between subgroups of G and subgroups of the quotient group.

Theorem (The Correspondence Theorem)**.** Let G be a group with $N \trianglelefteq G$. Then there is a bijection from $\{A \leq G : N \leq A\}$ to the set of all subgroups of G/N. In particular, every subgroup of $\overline{G} = G/N$ is of the form $A/N = \overline{A}$ for some subgroup A of G containing N. Namely, the complete pre-image of the subgroup of G/N under the natural projection from G to G/N. This bijection has the following properties:

(i) $A \leq B$ if and only if $\overline{A} \leq \overline{B}$; that is, $A/N \leq B/N$.

(ii) If $A \leq B$, then $|B : A| = |B/N : A/N| = |\overline{B} : \overline{A}|$.

(iii) $\overline{\langle A, B\rangle} = \langle \overline{A}, \overline{B}\rangle$.

(iv) $\overline{A \cap B} = \overline{A} \cap \overline{B}$

(v) $A \trianglelefteq G$ if and only if $\overline{A} \trianglelefteq \overline{G}$ $(A/N \trianglelefteq G/N)$

Every time we have a bar over a group it represents the image of the natural projection. Now, the question is, "how do we construct the bijection form the subgroups of G containing N to the subgroups of G/N?". The answer is the projection map $\pi : G \to G/N$. Let $A \subseteq G/N$ where

$$\pi(A) = \{aN : a \in A\} \leq G/N.$$

That point here is that π will provide our bijection *this is for you to prove.* One can also prove (ii) on your own. The exam will consists of comp problems, using H and K, and one of the properties from the correspondence theorem.

9 Week 9

Important theorems (that will come later):

Theorem (Cauchy's Theorem). If G is a finite group and p is a prime dividing the order of G, then G has a element of order p.

Theorem (Sylow Theorem). If G is a finite group of order $p^\alpha \cdot m$ where p is a prime and p does not divide m, then G has a subgroup of order p^α where $\alpha \in \mathbb{N}$.

In the following, we will relax one of the assumptions found in Cauchy's theorem and assume that it is abelian.

Proposition (21 Dummit and Foote). If G is a finite abelian group and p is a prime dividing $|G|$, then G contains an element of order p.

Proof. We will use strong induction to prove this result. (we will assume the result holds for all groups with order is less than $|G|$ and show this implies the result for $|G|$). Since $|G| > 1$, there exists an element $x \in G$ such that $x \neq 1_G$. If $|G| = p$, then we are done by Lagrange's theorem; that is, any group with order prime p contains an element of order p. Assume that $|G| > p$. Suppose that $p \mid |x|$. Hence,

$$|x| = p \cdot n \quad \text{for some } n \in \mathbb{Z}.$$

We know that $|x^n| = \frac{|x|}{(|x|,n)} = \frac{p \cdot n}{n} = p$. Again, we have an element of order p and we are done. We now assume that $p \not| |x|$. Let $\langle x\rangle = N$. Since G is abelian, this N is normal in G. Hence, Lagrange's theorem implies that, we can say that

$$|G : N| = \Big|G/N\Big| = \frac{|G|}{|N|}.$$

Since $N \neq 1_G$, we have that

$$\frac{|G|}{|N|} < |G|.$$

Since $p \not| |N|$, we have that $p \mid |G/N|$. By our inductive assumption, we can conclude that there exists a $yN \in G/N$ such that

$$|yN| = p.$$

Since $yN \neq 1N$, we have $y \notin N$ (in order for $y \in N$ we would need to raise to p in order for it to be the identity). But $y^p \in N$ since $(yN)^p = y^pN = 1_GN$. Since $\langle y^p\rangle \leq N$, we have that $\langle y\rangle \neq \langle y^p\rangle$.

That is, $\langle y^p \rangle < \langle y \rangle$. Furthermore, $|y| > |y^p|$. We know $|y^p| = \frac{|y|}{(|y|,p)}$. But since $p \mid |y|$, we have

$$|y^p| = \frac{|y|}{p}.$$

Thus, $p \mid |y|$ and we are done by our first case. Indeed, suppose for contradiction that $(|y|, p) = 1$, then from the above $|y^p| = |y|$ but this contradicts the assumption that $|y^p| < |y|$. This completes the induction and every finite abelian group with $p \mid |G|$ has an element of order p. ■

Central to this proof was finding an N normal in G. But what if we can't? The reason is because there are some groups which do not contain a normal subgroups.

Definition (Simple Groups). A (finite or infinite) group G is called **simple** if $|G| > 1$, and the only normal subgroups of G are just $\{1_G\}$ and G itself.

Example (Simple Groups). Z_p where p is prime.

The most important simple group (for us which will be proved later) is called the **Alternating group**. Consider $\sigma \in S_3$ and take $\sigma = (1\ 2\ 3)$ and so we can write this in a couple of ways:

$$\sigma = (1\ 3)(1\ 2) = (1\ 2)(1\ 3)(1\ 2)(1\ 3).$$

Notice that in general for any m-cycle in S_n, we have

$$\begin{aligned}&(a_1\ a_2\ \dots\ a_m)\\ &= (a_1\ a_m)(a_1\ a_{m-1})\dots(a_1\ a_3)(a_1\ a_2).\end{aligned}$$

That is, every $m-$cycle in S_n can be written as a product of 2-cycles (which are called transpositions). We know that every permutation in S_n can be written as a product of disjoint cycles. Hence, every permutation in S_n can be written as a product of transpositions; that is,

$$\langle T \rangle = S_n$$

where $T = \{(i\ j) : 1 \le i < j \le n\}$. As an example, in S_4, T has the elements

$$(1\ 2), (1\ 3), (1\ 4), (2\ 3), (2\ 4), (3\ 4).$$

Definition (Even (Odd) cycles). A permutation $\alpha \in S_n$ is called **even** if it can written as a product of an even number of transpositions. Otherwise, α is called **odd**.

Recall

$$(1\ 2\ 3) = (1\ 3)(1\ 2)$$

and so $(1\ 2\ 3)$ is even but we also know that

$$(1\ 2\ 3) = (1\ 2)(1\ 3)(1\ 2)(1\ 3).$$

Is this a well-defined definition? That is, can a permutation be both even and odd? The answer is **NO**. Permutations in S_n are either even or odd but not both.

Definition. For each $\sigma \in S_n$ define

$$\varepsilon(\sigma) = \begin{cases} 1 & \text{if } \sigma \text{ is even} \\ -1 & \text{if } \sigma \text{ is odd.} \end{cases}$$

Notice that ε defines a mapping from S_n to the multiplicative group $G = \{-1, 1\}$ (a subgroup of the quaternion group). Let $\sigma, \tau \in S_n$. Assume σ, τ are both even or both odd. Then $\sigma\tau$ is an even permutation and so $\varepsilon(\sigma\tau) = 1$ and $\varepsilon(\sigma)\varepsilon(\tau) = 1$. If one of σ and τ is even and the other is odd then $\sigma\tau$ is odd and so $\varepsilon(\sigma\tau) = -1$ and $\varepsilon(\sigma)\varepsilon(\tau) = -1$. And so ε defines a homomorphism. By the first isomorphism theorem, we have

$$S_n/\ker(\varepsilon) \cong \varepsilon(S_n) = \{1, -1\}.$$

By Lagrange's theorem,

$$\begin{aligned}\frac{|S_n|}{|\ker(\varepsilon)|} = |\{1, -1\}| \Longrightarrow \frac{n!}{|\ker(\varepsilon)|} &= 2 \\ \Longrightarrow \frac{n!}{2} &= |\ker(\varepsilon)|.\end{aligned}$$

That is,

$$\begin{aligned}\ker(\varepsilon) &= \{\sigma \in S_n : \varepsilon(\sigma) = 1\} \\ &= \{\sigma \in S_n : \sigma \text{ is even}\}.\end{aligned}$$

Definition (The Alternating Group is the Kernel of Some homomorphism)**.** The alternating group of degree n, denoted by A_n is the kernel of homomorphism ε, but it is more common referred to as the set of all even permutations in S_n.

9.1 Opening Remarks

Remark (An Alternative Proof to Problem 1 (a) on Homework)**.** If $\varphi : G \to J$ where φ is surjective. If J is abelian, then any subgroup of G which contains $\ker(\varphi)$ is normal.

Proof. Note that the first isomorphism theorem says that

$$G/\ker(\varphi) \cong \varphi(G)$$

Also, $\varphi(G) = J$ since φ is surjective. So, $G/\ker(\varphi) \cong J$ and hence, $G/\ker(\varphi)$ is abelian. Since J is abelian, every subgroup of $G/\ker(\varphi)$ is normal. Hence, by the 4th iso theorem every subgroup of G containing $\ker(\varphi)$ is also normal. ■

Remark (The Correspondence Theorem)**.** Every subgroup of G/N is the image of exactly one subgroup of G which contains N. That is, the subgroup of G/N is the image of exactly one subgroup of G containing N, through the natural projection.

This is to say that if you have a subgroup of G/N, it is of the form

$$\pi(A) = \{aN : a \in A\}$$

where $A \leq G$ which contains N.

9.2 Group Actions

Let G be a group and A be a nonempty set. A group action is a mapping $G \times A \to A$ denoted as

$$g \cdot a \ \ \forall g \in G,\ a \in A.$$

Recall for each $g \in G$, there exists a mapping $\sigma_g : A \to A$ such that $g \mapsto g \cdot a$ which is a permutation or an element of the symmetric group S_A on A. We defined $\varphi : G \to S_A$ defined by $g \mapsto \sigma_g$. We proved that this φ is a homomorphism. We called it the permutation representation of G on A. Recall further the following important facts:

(1) The kernel of the action is
$$\{g \in G : g \cdot a = a \quad \forall a \in A\} = \ker(\varphi).$$

(2) Stabilizer of $a \in A$ in G
$$G_a = \{g \in G : g \cdot a = a\}.$$

(3) **New Fact:** An action is faithful if its kernel is $\{1_G\}$.

Example (Faithful Mapping)**.** We know that S_n acts on the set $A = \{1, 2, \ldots, n\}$ by $\varphi \cdot i = \sigma(i)$ for all $\sigma \in S_n$ and all $i \in A$ where $\varphi : S_n \to S_n$ is the identity map. This is true because $\sigma(i) = i$ for all $i \in A$ and so we have $\sigma = 1_{S_n}$ and so this action is faithful.

Recall that $|G_i| = (n-1)!$. In general, we can say that $\ker(\varphi) = \bigcap_{a \in A} G_a$. Given a homomorphism $\psi : G \to S_A$ where A is a nonempty set and G is a group, the mapping defined by
$$g \cdot a = \psi(g)(a)$$
is a group action. Indeed, we see that for all $g_1, g_2 \in G$, we have
$$\begin{aligned}(g_1 g_2) \cdot a &= \varphi(g_1 g_2)(a) \\ &= (\varphi(g_1) \circ \varphi(g_2)(a)) \\ &= \varphi(g_1)\Big(\varphi(g_2)(a)\Big) \\ &= \varphi(g_1)(g_2 \cdot a) \\ &= g_1 \cdot (g_2 \cdot a)\end{aligned}$$
and
$$1_G \cdot a = \varphi(1_G)(a) = 1_{S_A}(a) = a.$$

Notice that
$$\begin{aligned}\mathcal{O}_a &= \{b \in A : b \sim a\} \\ &= \{b \in A : b = g \cdot a \text{ for some } g \in G\} \\ &= \{g \cdot a : g \in G\}.\end{aligned}$$

Proposition (Orbit Stabilizer Theorem)**.** Let G be a group acting on a nonempty set A. The relation on A defined by $a \sim b$ if and only if $a = g \cdot b$ for some $g \in G$ is an equivalence relation. For each $a \in A$, the number of elements in the equivalence class of containing a is the index $|G : G_a|$; that is, the number of left-cosets of G_a in G. We denoted the notation for the orbit of a as $\mathcal{O}_a$.

Proof. To prove the last statement, we need to construct a mapping that is bijective between $\mathcal{O}_a$ between

Let $\mathcal{O}_a$ be the orbit of G containing a. Let $b \in \mathcal{O}_a$. Then $b = g \cdot a$ for some $g \in G$. Notice that gG_a is a left-coset of G_a. Define $f : \mathcal{O}_a \to G/G_a$ by $b = g \cdot a \longmapsto gG_a$. Note that we have to be careful here because we might not have a well-defined mapping here. Nevertheless, let us prove that this mapping is onto. Let $gG_a \in G/G_a$. Then $g \cdot a \in \mathcal{O}_a$ and $g \cdot a = gG_a$ and so f is surjective. Notice $f(g \cdot a) = f(h \cdot a)$. Then we have
$$\begin{aligned}& gG_a = hG_a \\ & \implies h^{-1}g \in G_a \\ & \implies (h^{-1}g) \cdot a = a \\ & \implies h \cdot (h^{-1}g) \cdot a) = h \cdot a \\ & \implies (hh^{-1}g) \cdot a = h \cdot a \\ & \implies g \cdot a = h \cdot a.\end{aligned}$$

Hence, f is injective. But this tells us that f is indeed well-defined! Thus, f is a bijection from $\mathcal{O}_a$ to G/G_a. Thus, $|\mathcal{O}_a| = |G : G_a|$. ■

Example. Let $A = \{1, 2, 3\}$ and let $\sigma \in S_n$. We know that S_n acts on $A = \{1, 2, \ldots, n\}$. Given $i \in A$, find $\mathcal{O}_i = \{\sigma(i) : \sigma \in S_n\} = A$. What does the Orbit Stabilizer Theorem say in this example? It tells us that

$$|\mathcal{O}_i| = |S_n : G_i| \Longrightarrow n = \frac{|S_n|}{|G_i|} = \frac{n!}{|G_i|} \Longrightarrow |G_i| = \frac{n!}{n} = (n-1)!.$$

10 Week 11

10.1 Caley's Theorem

Last time we discussed G acting on itself by left-multiplication which is a special case of G acting on the cosets of a subgroup H by left-multiplication; that is, $g \cdot aH = (ga)H$ for all $g \in G$ and all $aH \in G/H$. We will prove some things using the permutation representation of g acting on the cosets of a subgroup H. That is,

$$\varphi : G \to S_{G/H}$$

defined by $\psi(g) = \sigma_g$ where $\sigma_g(xH) = g \cdot xH = (gx)H$ is a homomorphism.

Theorem. Let G and be a group and $H \leq G$. Let G act on $A = G/H$ by left multiplication. Let $\pi_H : G \to S_{G/H}$ be the associated permutation representation afforded by this action. Then

(1) G acts transitively on A; that is, for any $aH, bH \in A$, there exist a $g \in G$ such that

$$bH = g \cdot aH$$

(2) $G_{1H} = H$.

(3) The kernel of the action $(\ker(\pi_H))$ is

$$\bigcap_{x \in G} xHx^{-1}$$

and the kernel of π_H is the largest normal subgroup of G contained in H.

Proof. (a) To see that G acts transitively on A, let $aH, bH \in A$. Then let $g = ba^{-1}$ and so

$$\begin{aligned} g \cdot aH &= (ba^{-1}) \cdot aH \\ &= (ba^{-1})aH \\ &= b(a^{-1}a)H \\ &= bH. \end{aligned}$$

(b) Note that

$$\begin{aligned} G_{1H} &= \{g \in G : g \cdot 1H = 1H\} \\ &= \{g \in G : (g1)H = H\} \\ &= \{g \in G : gH = H\} \\ &= \{g \in G : 1^{-1}g \in H\} \\ &= \{g \in G : g \in H\} \\ &= H. \end{aligned}$$

(c) By definition,

$$\begin{aligned} \ker(\pi_H) &= \{g \in G : \pi_H(g) = 1_{S_{G/H}}\} \\ &= \{g \in G : g \cdot xH = xh \ \forall xH \in A\} \\ &= \{g \in G : gxH = xH \ \forall x \in G\} \\ &= \{g \in G : x^{-1}gx \in H \ \forall x \in G\} \\ &= \{g \in G : g \in xHx^{-1} \forall x \in G\} \\ &= \bigcap_{g \in G} xHx^{-1}. \end{aligned}$$

Observe that $\ker(\pi_H) \trianglelefteq G$ and $\ker(\pi_H) \leq H$. Let $N \trianglelefteq G$ such that $N \leq H$ if and only if $xNx^{-1} \subseteq xHx^{-1}$. Then

$$N = xNx^{-1} \leq xHx^{-1} \quad \forall x \in G.$$

Hence,

$$N \leq \bigcap_{x \in G} xHx^{-1} = \ker(\pi_H).$$

■

Theorem (Caley's Theorem)**.** Every group is isomorphic to a subgroup of some symmetric group. In particular, if $|G| = n < \infty$, then G is isomorphic to a subgroup of S_n.

Proof. Let $H = \{1\}$ and by the first iso theorem, we have that

$$G/\ker(\pi_H) \cong \pi_H(G) \leq S_G.$$

Since $\ker(\pi_H) \leq H$, we have

$$\ker(\pi_H) \leq \{1\} \implies \ker(\pi_H) = 1.$$

Hence,

$$G \cong G/\ker(\pi_H) \cong \pi_H(G).$$

where $\pi_H(G) \leq S_G$. ■

The permutation representation of this action when $H = \{1\}$ is called the left regular representation of G.

Corollary (Classifying Groups)**.** If G is a finite group of order $n \in \mathbb{N}$ and p is the smallest prime dividing n, then any subgroup of index p is normal in G.

Proof. Suppose $H \leq G$ and $|G : H| = p$. Let π_H be the permutation representation afforded by G

acting on the left cosets of H by left-multiplication. Let $K = \ker(\pi_H)$ and $|H : K| = k$. Notice that $k \mid |H|$ (by Lagrange's Theorem) and so $k \mid |G|$. Then by our tower laws, we see that

$$|G : K| = |G : H||H : K| = pk.$$

Since H has p cosets in G,

$$\pi_H : G \to S_p.$$

By the first isomorphism theorem,

$$G/K \cong \pi_H(G) \leq S_p.$$

This implies that $pk = |G : K| = \frac{|G|}{|K|}$. Hence,

$$pk = |\pi_H(G)|$$

and so

$$pk \mid |S_p| \implies pk \mid p^! \implies k \mid (p-1)!.$$

However, since $k \mid |G|$ any prime factor of k. However, since $k \mid |G|$ any prime factors of k would be larger than or equal to p, so $k = 1$ minimality of p. Indeed, suppose otherwise. Then $q \mid k$ implies $q \mid |G|$ and $q \mid (p-1)!$. But this tells us that since $|H : K| = k$, we have $|H : K| = 1$ and so $H = K$. Hence, $H = K \trianglelefteq G$. ∎

Example. Given a group G of order 15 by Cauchy's Theorem, we know that G contains a subgroup of order 5, and by the previous corollary, we know that subgroup is normal. Since its index must be 3 which is the smallest prime dividing 15.

10.2 Groups Acting on Themselves

Groups acting on themselves by conjugation (class equation). Let G be a group acting on itself by conjugation. Define $g \cdot a = gag^{-1}$ for all $g \in G$ and all $a \in G$. We proved that this is a group action.

Definition. We say $a, b \in G$ are conjugate in G if there exists a $g \in G$ such that

$$b = gag^{-1};$$

that is, a, b are conjugate in G if and only if $a, b \in \mathcal{O}_x$ for some $x \in G$. Recall that

$$\mathcal{O}_x = \{g \cdot x : g \in G\} = \{gxg^{-1} : g \in G\}.$$

The orbits of G acting on itself by conjugation are called the **conjugacy classes** of G.

Example.
- If G is abelian, then G acting on itself by conjugation is the **trivial action**; that is, $g \cdot a = gag^{-1} = a$ for all $a \in G$ and $g \in G$.

This tells us that the conjugacy class of a in G is

$$\begin{aligned}\{gag^{-1} : g \in G\} &= \{a : g \in G\} \\ &= \{a\}.\end{aligned}$$

More generally, for G acting on itself where G is not assumed to be abelian, the conjugacy class of some $g \in G$ is just $\{g\}$ if and only if $g \in Z(G)$.

Example. If $|G| > 1$, then G does not act transitively on itself by conjugation. (If transitive, there is only one orbit) The conjugacy class of 1_G is just $\{1_G\}$. That is,

$$\{g1g^{-1} : g \in G\} = \{1 : g \in G\} = \{1\}.$$

Note that also the action of conjugation can be generalized to G acting on $\mathcal{P}(G)$ by

$$g \cdot S = gSg^{-1} \ \forall g \in G,\ S \in \mathcal{P}(G)$$

where $\mathcal{P}$ is the power set of G.

Definition (Conjugate). Two subsets S and T of G are called **conjugate** in G if there exists a $g \in G$ such that $S = gTg^{-1}$.

Lemma (Result Needed to Prove Class Equation (Prop 6)). The number of conjugates of a subset S in a group G is $|G : N_G(S)|$. In particular, the number of conjugates of an element $s \in G$ is $|G : C_G(s)|$.

Proof. Let $S \subseteq G$. Using Orbit-stabilizer theorem, we know that the conjugacy class of S equals $|G : G_s|$. For this particular action,

$$\begin{aligned} G_s &= \{g \in G : g \cdot S = S\} \\ &= \{g \in G : gSg^{-1} = S\} \\ &= N_G(S). \end{aligned}$$

The second statement follows from

$$\begin{aligned} N_G(\{s\}) &= \{g \in G : g\{s\}g^{-1} = \{s\}\} \\ &= \{g \in G : gsg^{-1} = s\} \\ &= C_G(\{s\}). \end{aligned}$$

That is, $N_G(s) = C_G(s)$. ■

Theorem (The Class Equation). Let G be a finite group and let $g_1, g_2, \ldots, g_r$ be representatives of distinct distinct conjugacy classes of G not contained in $Z(G)$. Then

$$|G| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|.$$

Proof. We claim that $z \in G$ has conjugacy class $\{z\}$ if and only if $z \in Z(G)$. You can show this in your free time. Let

$$Z(G) = \{1, z_2, z_3, \ldots, z_m\}.$$

Let $k_1, k_2, \ldots, k_r$ be the conjugacy classes of G not contained in $Z(G)$ and let g_i be a representative of k_i (element of) for each i. Then the full set of conjugacy classes is

$$\{1\}, \{z_2\}, \ldots, \{z_m\}, k_1, k_2, \ldots, k_r.$$

Since these partition G, we have

$$\begin{aligned}|G| &= \sum_{i=1}^{m} 1 + \sum_{i=1}^{r} |K_i| \\ &= |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|\end{aligned}$$

by the previous proposition. ∎

Example. (1) If G is abelian of order n, then the class equation tells us nothing; that, is

$$|G| = |Z(G)|.$$

(2) Calculate the class equation of Q_8. Note that in any group $\langle x \rangle \leq C_G(x)$. We know that $i \notin Z(G)$ (to see this compute i and j together in different order). We also know that $|\langle i \rangle| = 4$ which implies that $|Q_8 : \langle i \rangle| = 2$. Also, we know that $i \notin Z(Q_8)$. This implies that the $C_{Q_8}(i) < Q_8$ (proper subgroup of Q_8). (Recall that $C_G(x) = G \iff x \in Z(G)$). So, we have

$$2 = |Q_8 : \langle i \rangle| = |Q_8 : C_{Q_8}(i)||C_{Q_8}(i) : \langle i \rangle|$$

If $|Q_8 : C_{Q_8}| = 1$, then $Q_8 = C_{Q_8}(i)$ is a contradiction because we know that $i \notin Z(G)$. Thus, $|C_{Q_8}(i) : \langle i \rangle| = 1$ and $|Q_8 : C_{Q_8}(i)| = 2$. Thus, see that $C_{Q_8} = \langle i \rangle$. Now, it follows that the conjugacy class of i contains an index of 2 elements, namely, i and $-i$. Similarly, we can perform the same process to find all the conjugacy classes of Q_8,

$$\{1\}, \{-1\}, \{\pm i\}, \{\pm j\}, \{\pm k\}.$$

Thus, the class equation of Q_8 is

$$|\{1\}| + |\{-1\}| + |\{\pm i\}| + |\{\pm j\}| + |\{\pm k\}| = 2 + 2 + 2 + 2 = 8$$

Theorem. If p is a prime and P is a group of order p^α with $\alpha \geq 1$. Then P has a non-trivial center; that is, $Z(P) \neq \{1_P\}$ or $|Z(P)| > 1$.

Proof. By the class equation,

$$|P| = |Z(P)| + \sum_{i=1}^{r} |P : C_P(g_i)|$$

where $g_1, g_2, \ldots, g_r$ are representatives of the distinct non-central conjugacy classes. By definition, $C_P(g_i) \neq P$. (Note that $C_P(g_i) = P$ if and only if $g_i \in Z(G)$) for $i = 1, 2, \ldots, r$. So, p divides the

$$|P : C_P(g_i)| = \frac{|P|}{|C_P(g_i)|} = \frac{p^\alpha}{|C_p(g_i)|} = p^\beta$$

where $0 < \beta \leq \alpha$. Since p also divides $|P|$ it follows that

$$p \mid \left(|P| - \sum_{i=1}^{r} |P : C_P(g_i)|\right)$$

equivalently, we have $p \mid |Z(P)|$. Hence, it follows that $Z(G)$ is non-trivial. ∎

Corollary. If $|P| = p^2$ where p is a prime, then P is abelian. More precisely, $P \cong Z_{p^2}$ or $P \cong Z_p \times Z_p$.

Proof. Since $Z(P) \neq 1$ from the previous theorem, it follows that $|Z(P)| \in \{p, p^2\}$ and so we have

$$|P/Z(P)| \in \{1, p\}.$$

In either case, P is abelian (see result from one of homeworks; that is, $G/Z(G)$ implies that G is abelian). By the above, P is abelian. If P has an element of order p^2. If P has an element of order p^2, P is cyclic and isomorphic to Z_{p^2}. If P contains no elements of order p^2, then every non-identity element of P has order p. Let $x \in P$ such that $x \neq 1$ and $y \in P - \langle x \rangle$ ($y \neq 1$ since $1 \in \langle x \rangle$). Notice that x and y both have order p. Since $|\langle x, y \rangle| > |\langle x \rangle| = p$, we have $|\langle x, y \rangle| = p^2$; that is, $\langle x, y \rangle = P$. Note since P is abelian, $\langle x, y \rangle = \{x^a y^b : a, b \in \mathbb{Z}\}$. Also, since $|x| = |y| = p$, $\langle x \rangle \times \langle y \rangle \cong Z_p \times Z_p$. Defining a map from $Z_p \times Z_p$ to P by $(x^a, y^b) \mapsto x^a y^b$ gives an isomorphism. Hence, $Z_p \times Z_p \cong P$. ■

11 Week 12

11.1 Opening Remarks

Recall that the powerful idea that we encountered from last week is the mapping $\pi_H : G \to S_{G/H}$ especially $\ker(\pi_H) = \bigcap_{g \in G} gHg^{-1} \leq H$. If G is a group and H is a subgroup with $|G : H| = p$ where p is the smallest prime dividing $|G|$. Then $H \trianglelefteq G$. Note that this powerful idea makes one of the problems fro the exam trivial. That is, recall that the problem was:

$|G| = p^3 q$ with $p > q$. If $H \leq G$ with $|H| = p^3$. Then $|G : H| = q$ which is the smallest prime which divides $|G|$, $H \trianglelefteq G$.

11.2 Conjugacy in S_n

Recall that the important idea is not the permutation representation but rather the conjugacy classes which come from G acting on itself by conjugation. Let $\sigma = (1\ 2\ 3) \in S_5$ and let $\tau = (3\ 5) \in S_5$. Then

$$\tau\sigma\tau^{-1} = (3\ 5)(1\ 2\ 3)(3\ 5) = (1\ 2\ 5)$$

and

$$\tau\sigma\tau^{-1} = (2\ 3)(1\ 2\ 3)(2\ 3)^{-1} = (1\ 3\ 2)$$

and

$$\begin{aligned}\tau\sigma\tau^{-1} &= (1\ 4\ 3\ 2)(1\ 2\ 3)(1\ 4\ 3\ 2)^{-1} \\ &= (4\ 1\ 2)\end{aligned}$$

Lemma. Let $\sigma = (a_1\ a_2\ \ldots\ a_k) \in S_n$ be a k-cycle, then for any $\tau \in S_n$, we have

$$\tau\sigma\tau^{-1} = (\tau(a_1), \tau(a_2), \tau(a_3), \ldots, \tau(a_k)).$$

Proof. Notice that since $\sigma(a_1) = a_2$, then $\tau\sigma\tau^{-1}(\tau(a_1)) = \tau\sigma(a_1) = \tau(a_2)$ Similarly, $\sigma(a_2) = a_3$, and so

$$\tau^{-1}(\tau(a_2)) = \tau(a_3).$$

In general, $\sigma(a_i) = a_{i+1}$ and so $\tau\sigma\tau^{-1}(\tau(a_i)) = \tau(a_{i+1})$. For $1 \leq i < k$. Finally $\sigma(a_k) = a_1$ and so

$$\tau\sigma\tau^{-1}(\tau(a_k)) = \tau(a_1).$$

Noting that all the $\tau(a_i)$ are distinct since τ is injective and all the a_i's are distinct, we have $\tau\sigma\tau^{-1}$

contains the k-cycle
$$(\tau(a_1), \tau(a_2), \dots, \tau(a_k)).$$
Let $b \in A = \{1, 2, \dots, n\}$. But $b \notin \{a_1, a_2, \dots, a_k\}$, then $\sigma(b) = b$ and it follows that $\tau\sigma\tau^{-1}(\tau(b)) = \tau(b)$; that is, we see that $\tau(b)$ is fixed. Hence,
$$\tau\sigma\tau^{-1} = (\tau(a_1), \tau(a_2), \dots \tau(a_k)).$$
■

Proposition. Let σ and τ be elements of S_n and suppose that σ has cycle decomposition
$$\sigma_1\ \sigma_2\ \sigma_3 \dots\ \sigma_i = (a_1\ a_2\ a_3 \dots a_{k_i}),$$
then $\tau\sigma\tau^{-1}$ has cycle decomposition
$$\tau_1\ \tau_2\ \dots \tau_j$$
where $\tau_i = (\tau(a_1)\ \tau(a_2)\ \dots \tau(a_{k_i}))$.

Proof. The proof to the above proposition follows from the fact that $\tau\sigma_1\sigma_2\tau^{-1} = (\tau\sigma_1\tau^{-1})(\tau\sigma_2\tau^{-1})$. ■

Example. Let $\sigma = (1\ 2)(3\ 4\ 5)(6\ 7\ 8\ 9)$ and let $\tau = (1\ 3\ 5\ 7)(2\ 4\ 6\ 8)$. Then
$$\tau\sigma\tau^{-1} = (3\ 4)(5\ 6\ 7)(8\ 1\ 2\ 9)$$

Note that if we change the cycle types then the resulting will retain the same cycle type.

Definition. (1) If $\sigma \in S_n$ is the product of disjoint cycles of lengths $n_1, n_2, n_3, \dots, n_r$ where $n_1 \leq n_2 \leq \dots \leq n_r$ (including 1-cycles), the integers $n_1, n_2, \dots, n_r$ are called the cycle type of σ.

(2) If $n \in \mathbb{Z}^+$, a partition of n is any non-decreasing sequence of positive integers whose sum is n.

The punchline here is that cycle types have a one-to-one correspondence with partitions of n **(prove it)**.

We have seen that conjugation preserves cycle type; that is, two elements of S_n are conjugate implies they have the same cycle type. Consider the converse of this statement; that is, if two elements of S_n have the same cycle type, then they are conjugates of each other. Note that
$$\sigma_1 = (3\ 4\ 7\ 2)(1\ 5\ 8)$$
and
$$\sigma_2 = (1\ 2\ 3\ 4)(5\ 6\ 7).$$
We can find τ such that $\tau\sigma_1\tau^{-1} = \sigma_2$. Define τ as $\tau(3) = 1, \tau(4) = 2, \tau(7) = 3, \tau(2) = 4, \tau(1) = 5, \tau(5) = 6, \tau(6) = 8$ and $\tau(8) = 7$. Then
$$\tau = (1\ 5\ 6\ 8\ 7\ 3)(2\ 4)$$

Proposition (Proposition 11 from the book)**.** Two elements of S_n are conjugate if and only if they have the same cycle type. The number of cycle types of S_n equals the number of partitions of n.

Note that from this we see that each cycle type generates some kind of equivalence class. The goal of next lecture is to show that A_n is a simple group.